Trig graphs

https://www.geogebra.org/m/UjjwuM8p

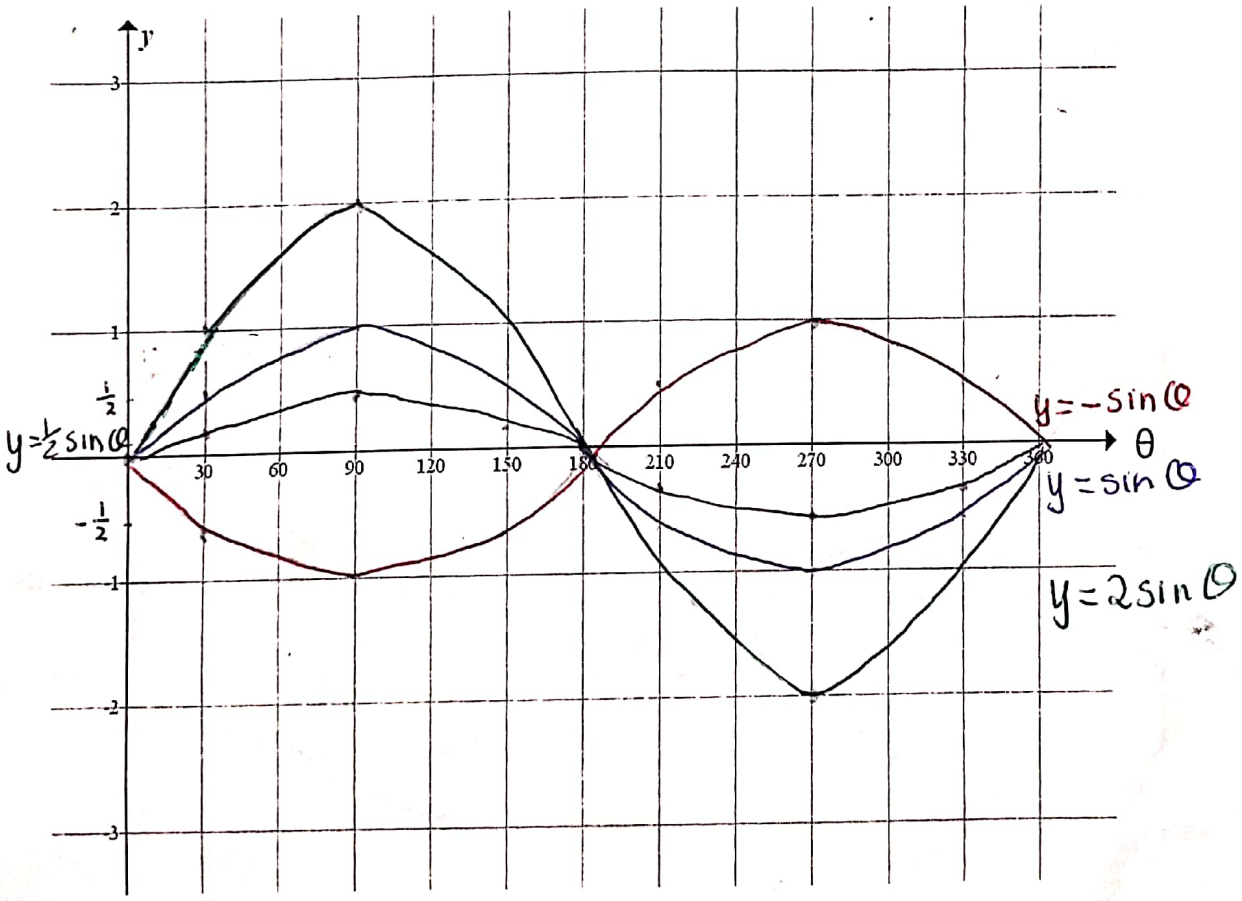
SINE GRAPH: $y = a\sin\theta + q$

Investigating the effect of a:

1. Complete the table below using a calculator:

θ	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\sin\theta$													
$-\sin\theta$													
$2\sin\theta$													

2. Now sketch $y = \sin\theta$, $y = -\sin\theta$ and $y = 2\sin\theta$ for the domain $0° \le \theta \le 360°$:



3. Now sketch $y = \frac{1}{2}\sin\theta$ on the above axes.
4. Describe the effect of a on the functions.

a stretches the graph and influences the max and min values: a affects reflection about the x-axis

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5. Complete the following table based on the sine graphs you have drawn:

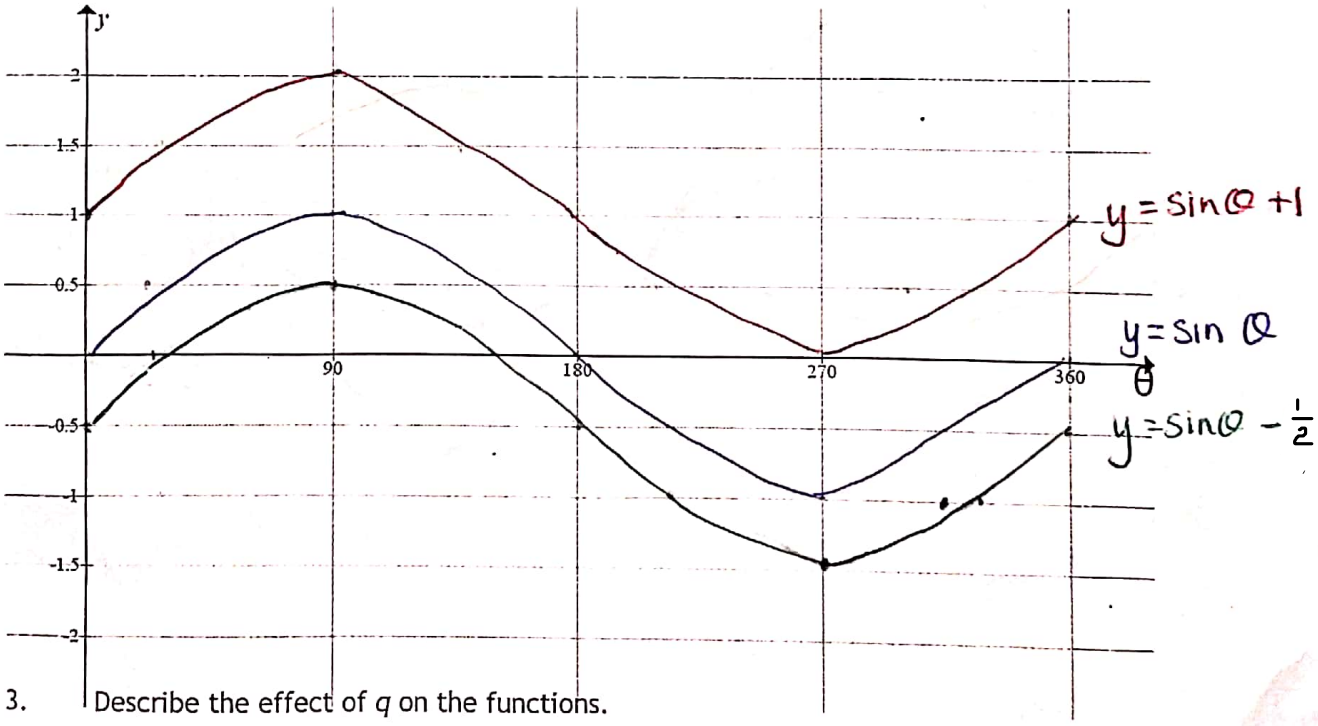
GRAPH	Amplitude	Maximum	Minimum	Period	Range
sin θ	1	1	-1	360	$y \in [-1;1]$
-sinθ	1	1	-1	360°	$y \in [-1;1]$
2 sin θ	2	2	-2	360°	$y \in [-2;2]$
0,5 sin θ	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	360°	$y \in [-\frac{1}{2};\frac{1}{2}]$

Investigating the effect of *q*:

1. Complete the table below using a calculator:

θ	0°	90°	180°	270°	360°
$\sin\theta$					
$\sin\theta+1$					
$\sin\theta-0,5$					

2. Now sketch $y=\sin\theta$, $y=\sin\theta+1$ and $y=\sin\theta-0.5$ for the domain $0° \le \theta \le 360°$:



3. Describe the effect of *q* on the functions.

If $q>0$, graph shifts up

If $q<0$, graph shifts down

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4. Complete the following table based on the sine graphs you have drawn:

GRAPH	Amplitude	Maximum	Minimum	Period	Range
sin θ	1	1	-1	360°	$y \in [-1; 1]$
sin θ + 1	1	2	0	360°	$y \in [0; 2]$
sin θ - 0,5	1	$\frac{1}{2}$	$-1\frac{1}{2}$	360°	$y \in [-1\frac{1}{2}; \frac{1}{2}]$

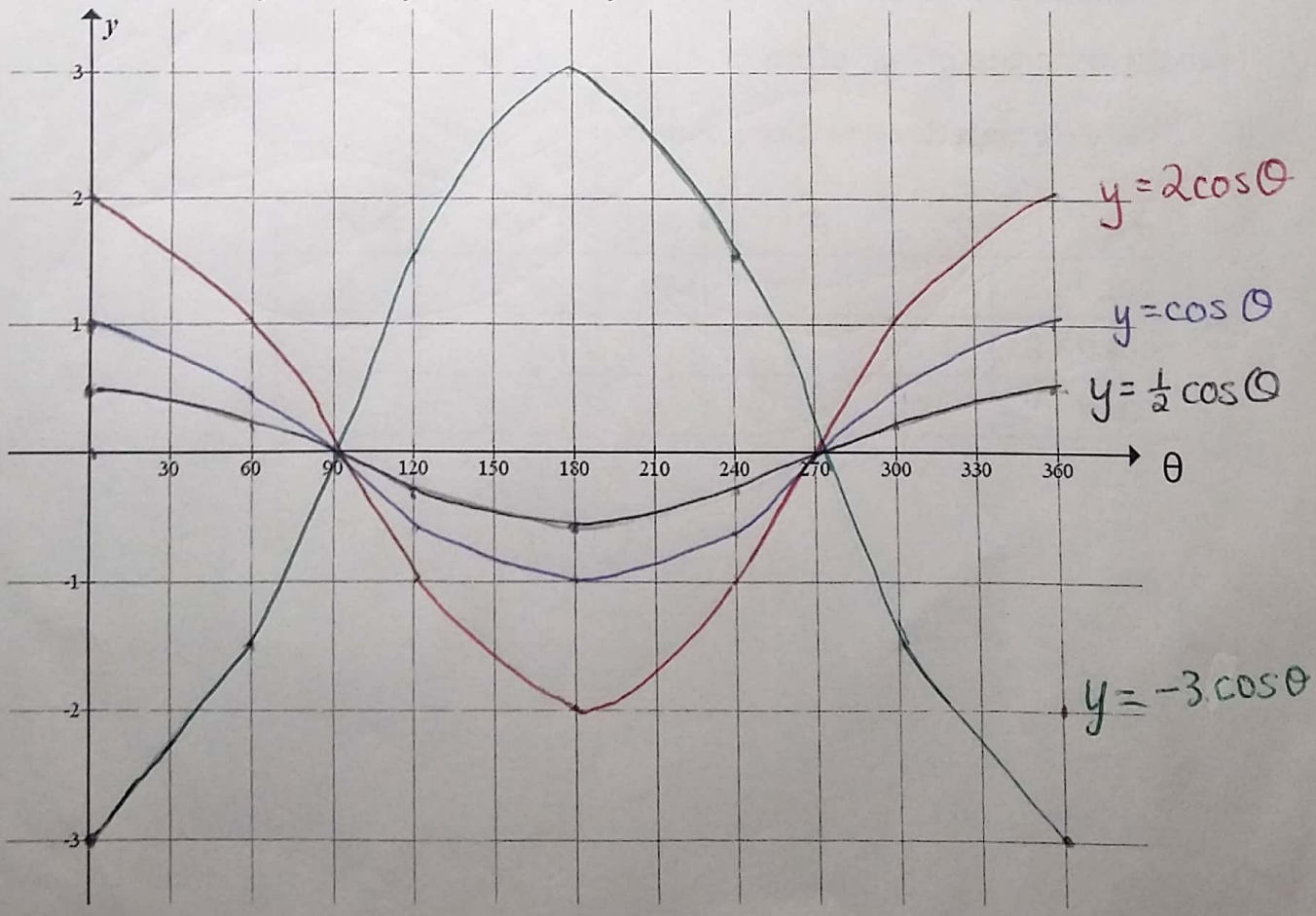
COSINE GRAPH: $y = a\cos\theta + q$

Investigating the effect of *a*:

1. Complete the table below using a calculator:

θ	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
$\cos\theta$													
$2\cos\theta$													
$-3\cos\theta$													

2. Now sketch $y = \cos\theta$, $y = 2\cos\theta$ and $y = -3\cos\theta$ for the domain $0° \le \theta \le 360°$:



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3. Now sketch $y = \frac{1}{2}\cos\theta$ on the axes.

4. Describe the effect of a on the functions.

a stretches the graph and influences the max and min. values. It affects the reflection about the x-axis

5. Complete the following table based on the cosine graphs you have drawn:

GRAPH	Amplitude	Maximum	Minimum	Period	Range
cos θ	1	1	−1	360°	$y \in [-1;1]$
2cos θ	2	2	−2	360°	$y \in [-2;2]$
-3cos θ	3	3	−3	360°	$y \in [-3;3]$
$\frac{1}{2}\cos\theta$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	360°	$y \in [-\frac{1}{2};\frac{1}{2}]$

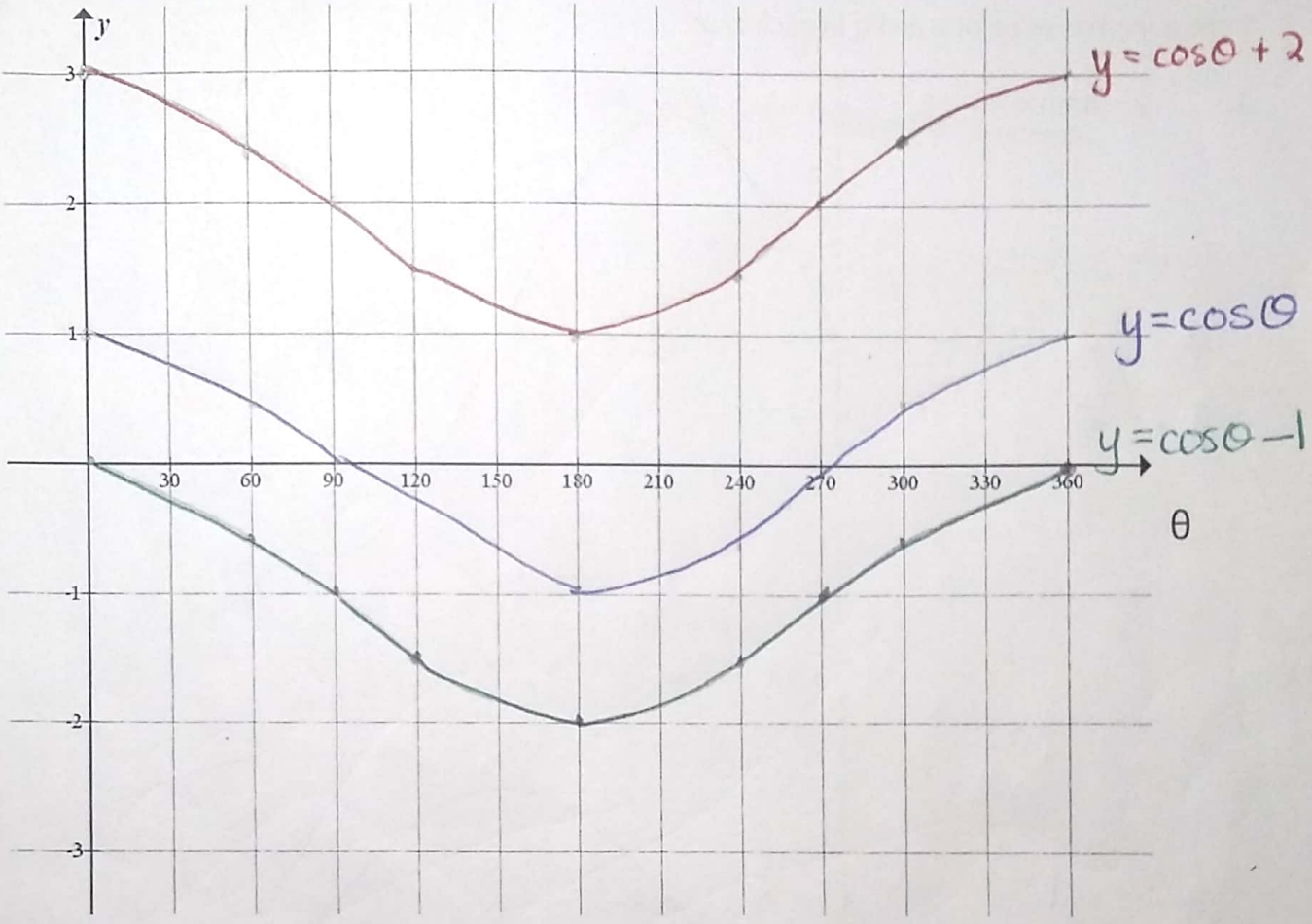
Investigating the effect of q:

1. Complete the table below using a calculator:

θ	0°	90°	180°	270°	360°
$\cos\theta$					
$\cos\theta+2$					
$\cos\theta-1$					

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2. Now sketch $y = \cos\theta$, $y = \cos\theta + 2$ and $y = \cos\theta - 1$ for the domain $0° \leq \theta \leq 360°$:



3. Describe the effect of q on the functions.

If $q > 0$, the graphs shifts up

If $q < 0$, the graph shifts down

4. Complete the following table based on the cosine graphs you have drawn:

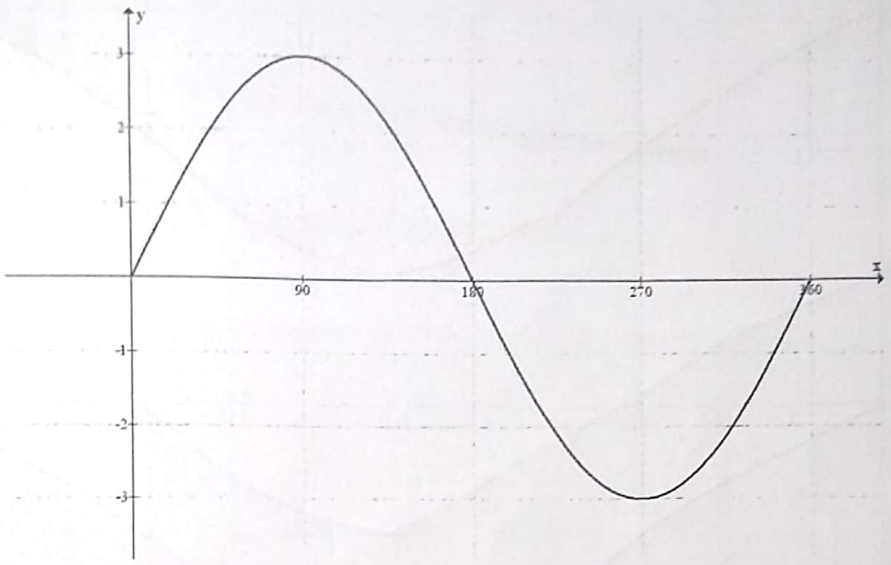
GRAPH	Amplitude	Maximum	Minimum	Period	Range
cos θ	1	1	-1	360°	$y \in [-1; 1]$
cos θ + 2	1	3	1	360°	$y \in [1; 3]$
cos θ - 1	1	0	-2	360°	$y \in [-2; 0]$

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Exercise 1:

Determine the value of a and q in each case:

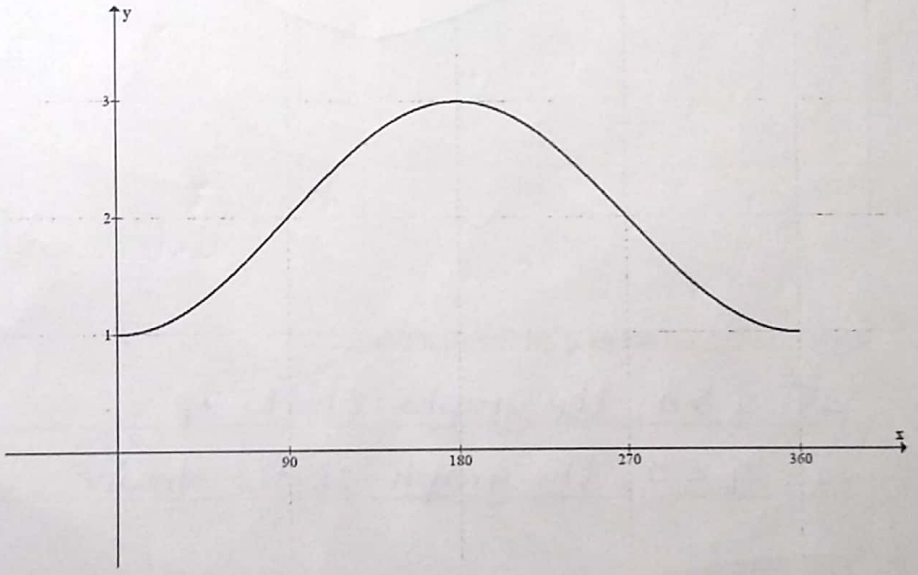
1. $y = a\sin x + q$



$a = 3$

$q = 0$

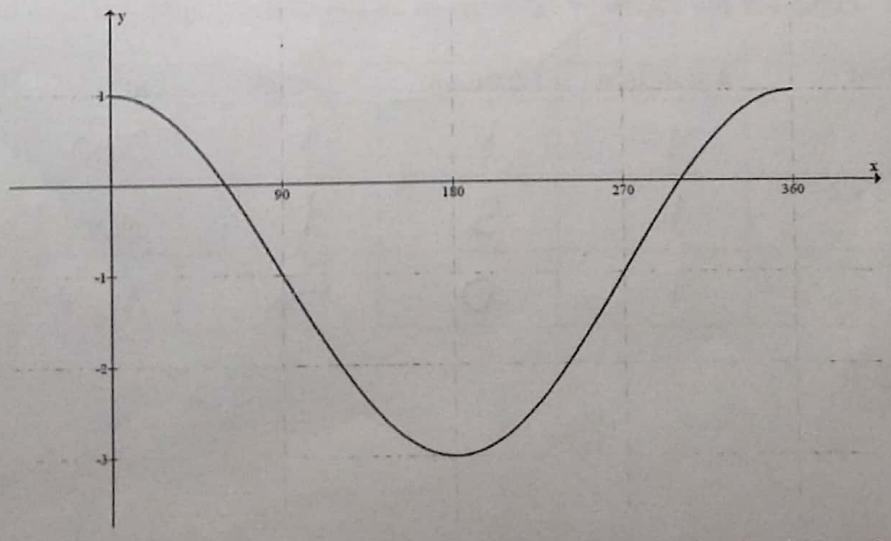
2. $y = a\cos x + q$



$a = -1$

$q = 2$

3. $y = a\cos x + q$



$a = 2$

$q = -1$

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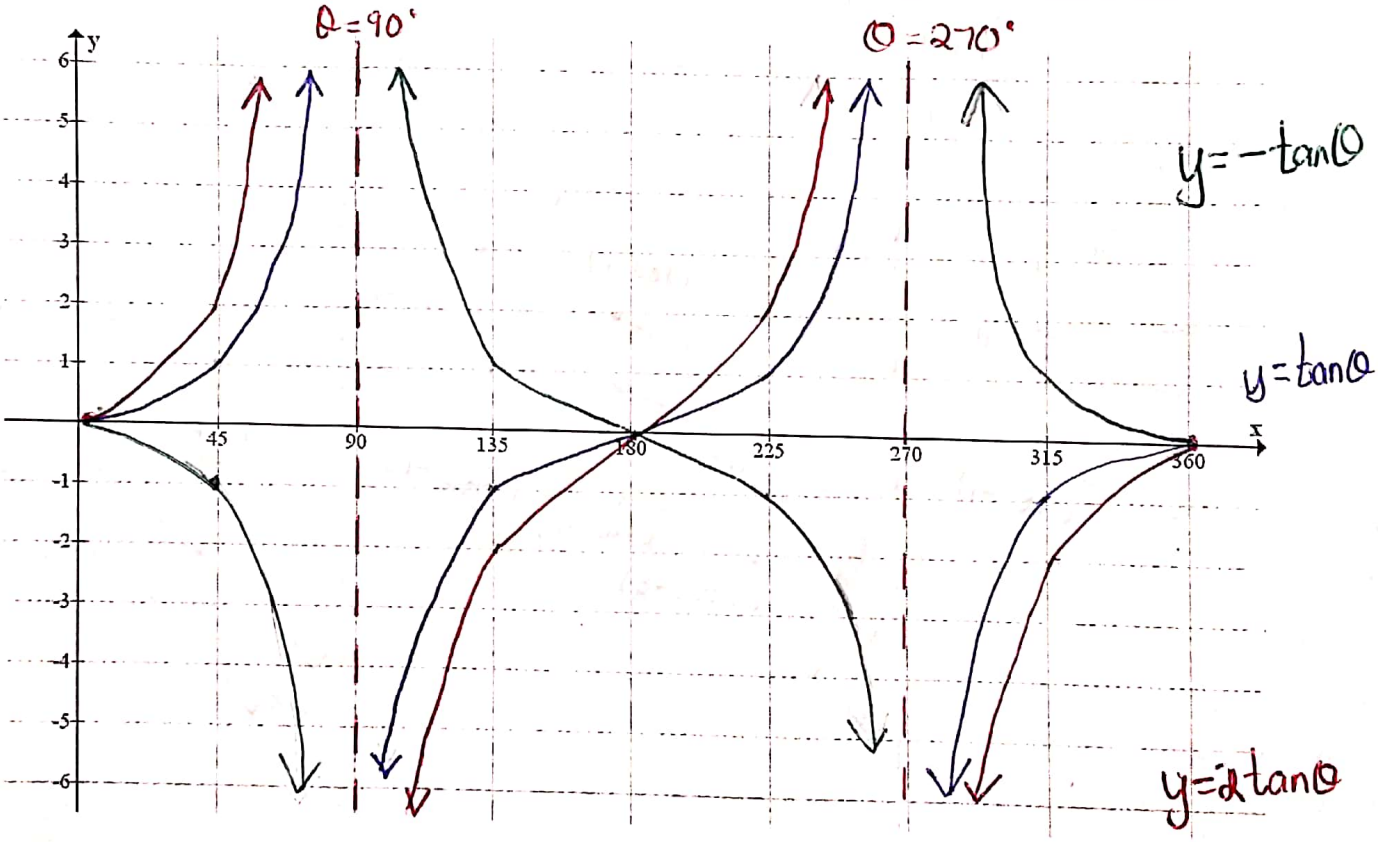
TANGENT GRAPH: $y = a\tan\theta + q$

Investigating the effect of *a*:

1. Complete the table below, with the help of a calculator:

θ	0°	45°	90°	135°	180°	225°	270°	315°	360°
$\tan\theta$									
$-\tan\theta$									
$2\tan\theta$									

2. Now sketch $y = \tan\theta$, $y = -\tan\theta$ and $y = 2\tan\theta$ for the domain $0° \le \theta \le 360°$:



3. Note that the tangent function completes **two full cycles** within 360°.

4. Complete the following table based on the tangent graphs you have drawn:

GRAPH	Amplitude	Maximun	Minimum	Period	Range
tan θ	—	—	—	180°	$y \in (-\infty : \infty)$
1 tan θ	—	—	—	180°	$y \in (-\infty : \infty)$
2 tan θ	—	—	—	180°	$y \in (-\infty : \infty)$

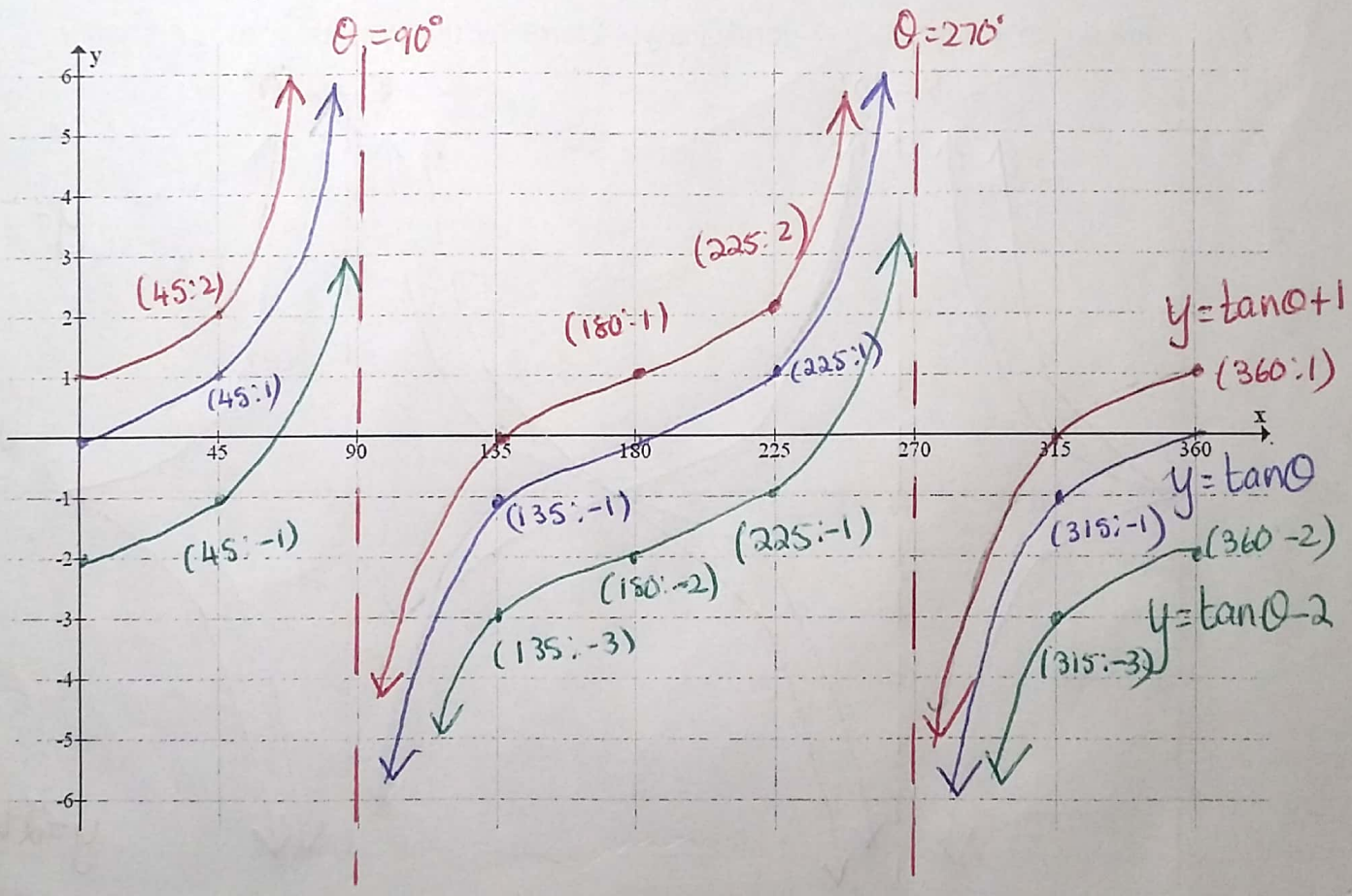
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Investigating the effect of q:

1. Complete the table below, with the help of a calculator:

θ	0°	45°	90°	135°	180°	225°	270°	315°	360°
$\tan\theta$									
$\tan\theta+1$									
$\tan\theta-2$									

2. Now sketch $y=\tan\theta$, $y=\tan\theta+1$ and $y=\tan\theta-2$ for the domain $0°\le\theta\le360°$:



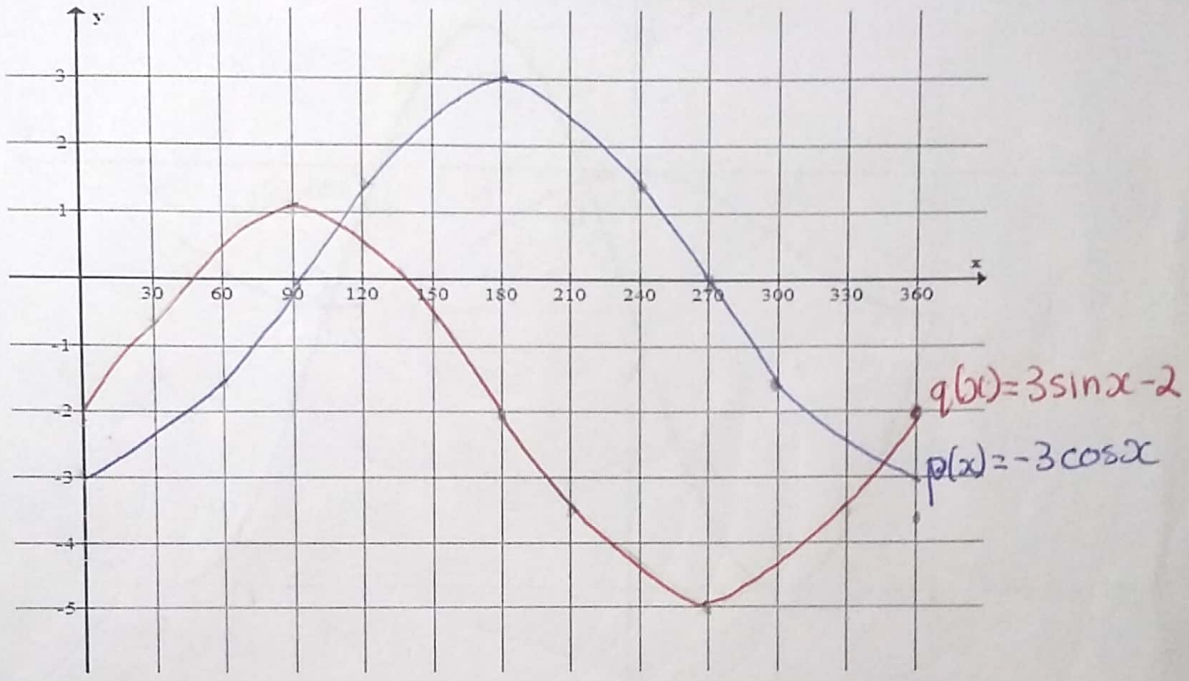
3. Complete the following table based on the tangent graphs you have drawn:

GRAPH	Amplitude	Maximum	Minimum	Period	Range
tan θ	–	–	–	180°	$y\in(-\infty;\infty)$
tan θ + 1	–	–	–	180°	$y\in(-\infty;\infty)$
tan θ - 2	–	–	–	180°	$y\in(-\infty;\infty)$

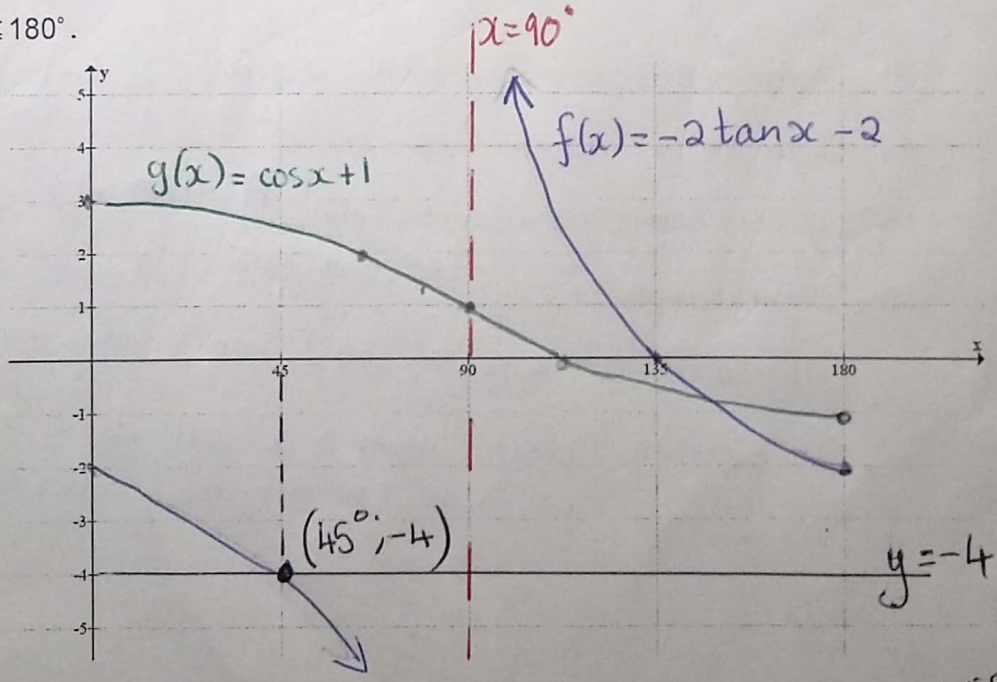
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Exercise 2

1. Sketch $p(x) = -3\cos x$ and $q(x) = 3\sin x - 2$ on the same set of axes for $x \in [0°; 360°]$ indicating the co-ordinates of all intercepts and turning points. Also show the co-ordinates of the endpoints.



2. Sketch $f(x) = -2\tan x - 2$ and g(x)=2cosx+1 on the same set of axes for $0° \le x \le 180°$.



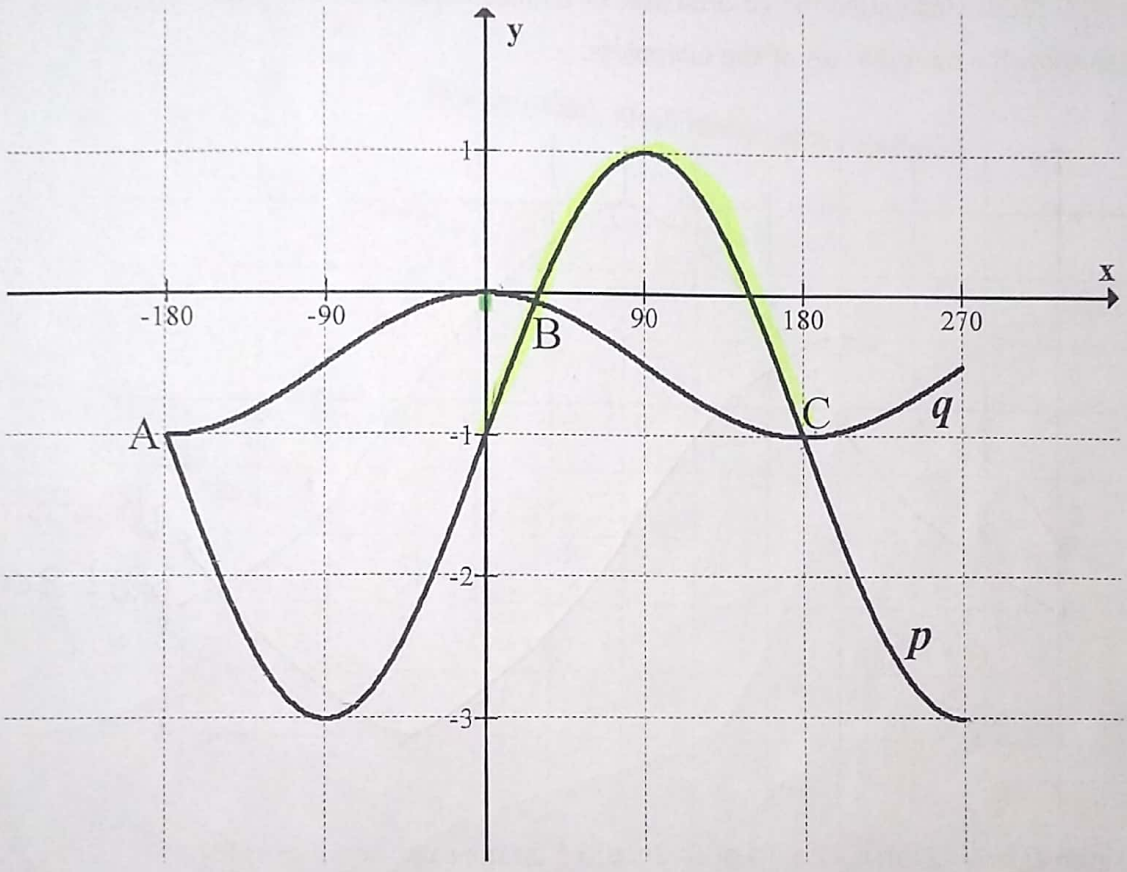
(a) Solve $f(x) = -4$, showing clearly on your graph how you read off your answer(s). $x = 45°$

(b) Write down the period of f and the period of g. period of f is 180°

Period of g(x) = 360°

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3. Given the graphs of $p(x) = a\sin x + b$ and $q(x) = c\cos x + d$ for $-180^\circ \le x \le 270^\circ$:



(a) Determine the values of a, b, c and d. $a = 2$; $b = -1$ $c = \frac{1}{2}$ $d = -1/2$

(b) Determine the co-ordinates of points A and C. $A(-180^\circ; -1)$ $C(180^\circ; -1)$

(c) Give two values of x for which $p(x) = q(x)$. -180° and 180°

(d) For which values of x is:

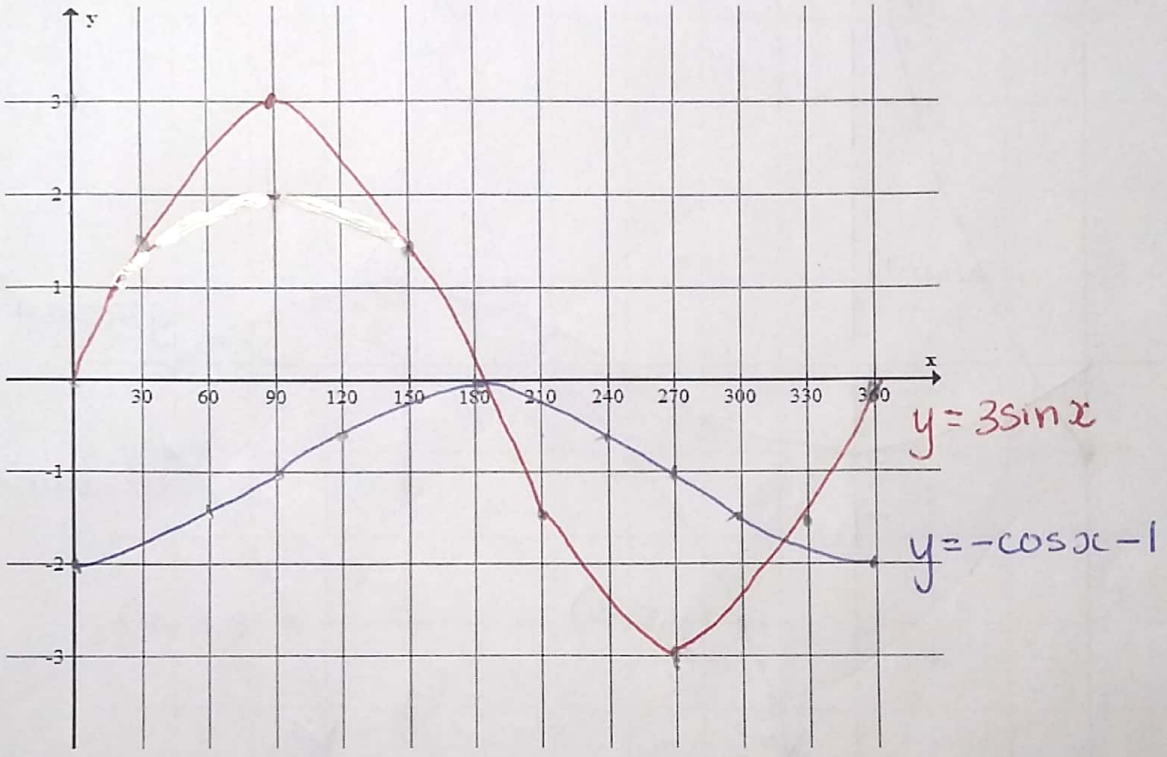
(1) $p(x) > -1$ $x \in (0^\circ; 180^\circ)$ See yellow area

(2) $q(x) \ge 0$ $q(x)$ is never > 0 but it is equal to 0 at the origin. ∴ where $x = 0^\circ$

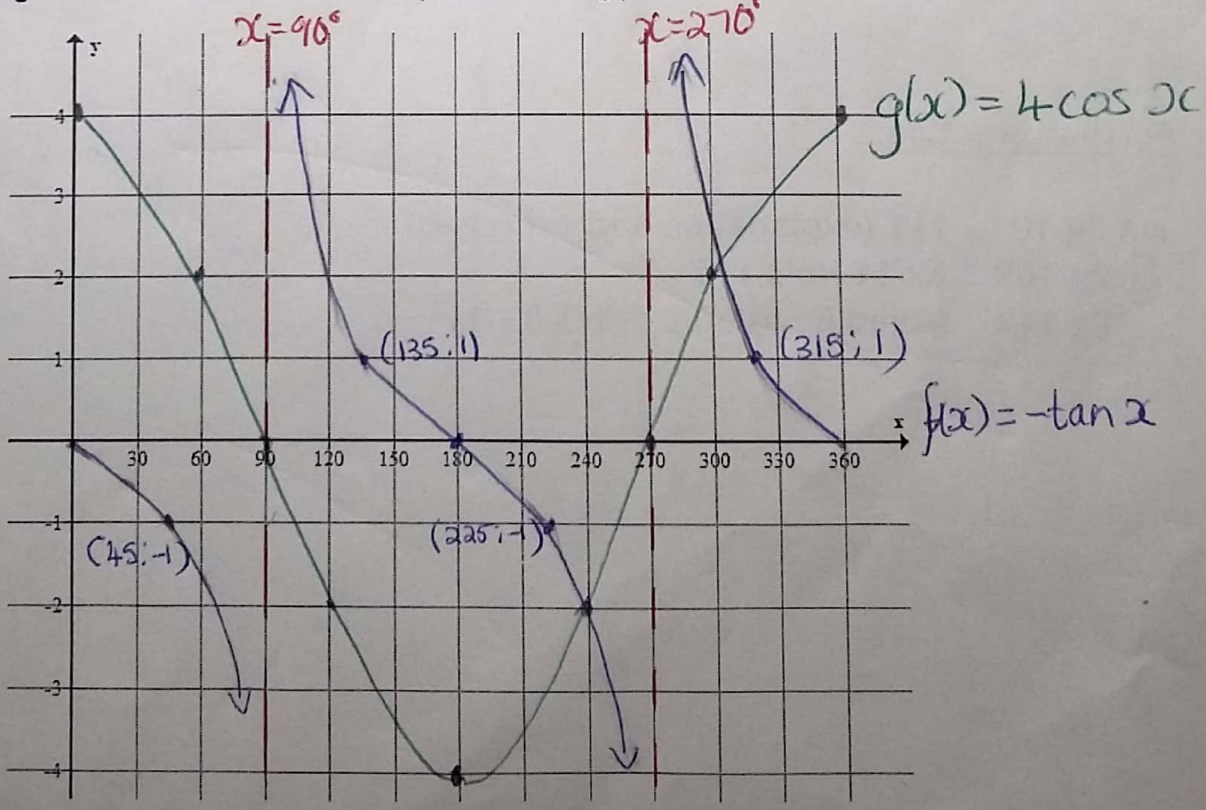
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Exercise 3:

1. Sketch $y = -\cos x - 1$ and $y = 3\sin x$ on the same set of axes for $x \in [0°;360°]$ indicating the co-ordinates of all intercepts and turning points.

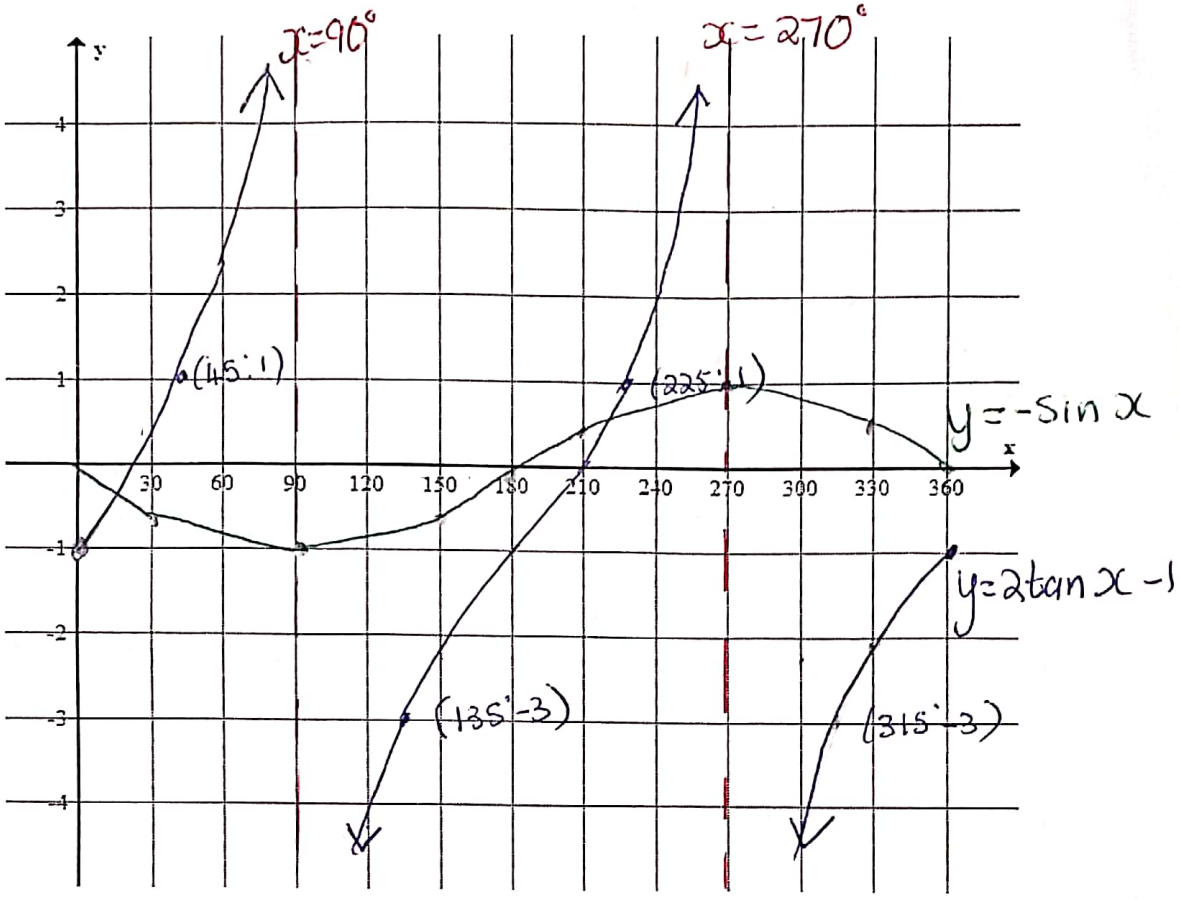


2. Sketch $f(x) = -\tan x$ and $g(x) = 4\cos x$ on the same set of axes for $x \in [0°;360°]$ indicating the co-ordinates of all intercepts and turning points.



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3. Sketch $y = 2\tan x - 1$ and $y = -\sin x$ on the same set of axes for $x \in [0°;360°]$ indicating the co-ordinates of all intercepts and turning points.



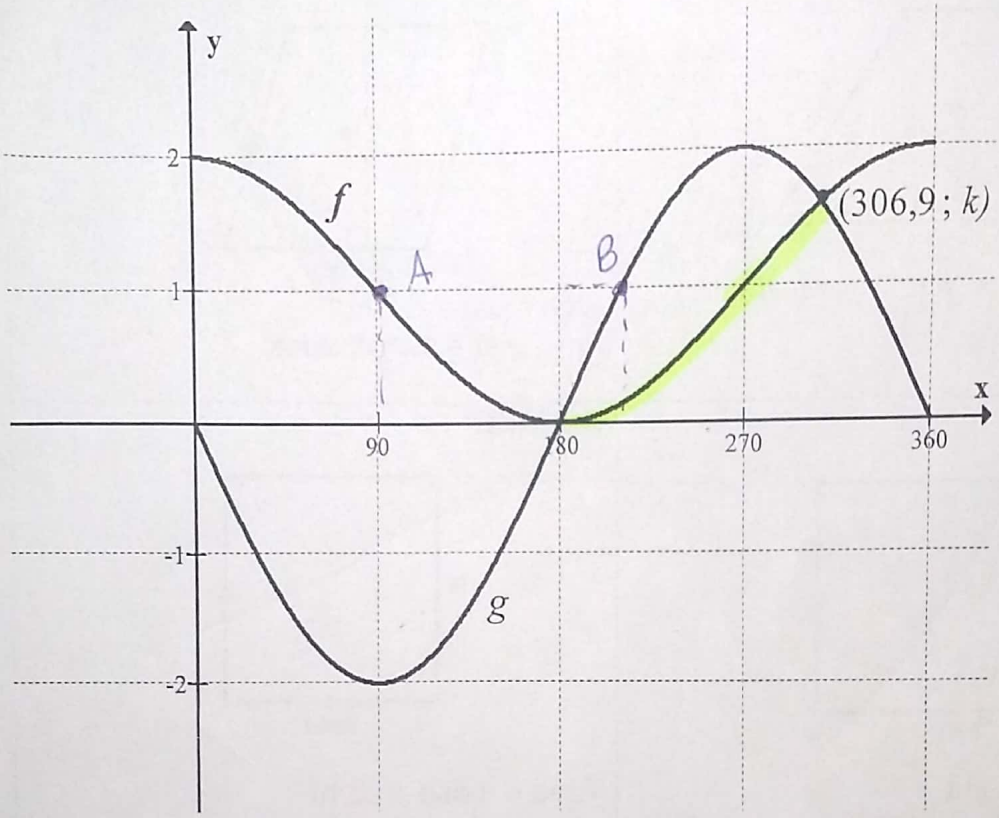
Further practice:

MA Pg 101 - 111 (explanation and exercises)
Pg 109 Ex 11 No's 1 -7
Pg 114 Mixed Revision Ex No's 8 - 11

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Exercise 4:

Given the graphs of $f(x) = a\cos x + b$ and $g(x) = c\sin x + d$ for $x \in [0°;360°]$:



a) Determine the values of a, b, c and d.

$a = 1$ $b = 1$

$c = -2$ $d = 0$

b) Determine the value of k to 2 decimal places.

EITHER $\cos(306,9) + 1 = 1,6$ OR $-2\sin(306,9) = 1,6$

c) Evaluate:

i) $f(90°) = 1$ (Read at A)

ii) $g(210°) = -2\sin(210) = 1$ (Read at B)

d) For which values of x is:

i) $f(x) = g(x)$ $x \in \{180°; 306,9°\}$

ii) $f(x) < g(x)$ $x \in (180°; 306,9°)$ See yellow section

iii) $f(x).g(x) \geq 0$ when y-values have same sign or are equal

or $+ \times +$ $- \times -$ $x \in [180°; 360°]$

e) Give two values of x for which $f(x) - g(x) = 2$.

$x = 0°$ and $x = 360°$

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